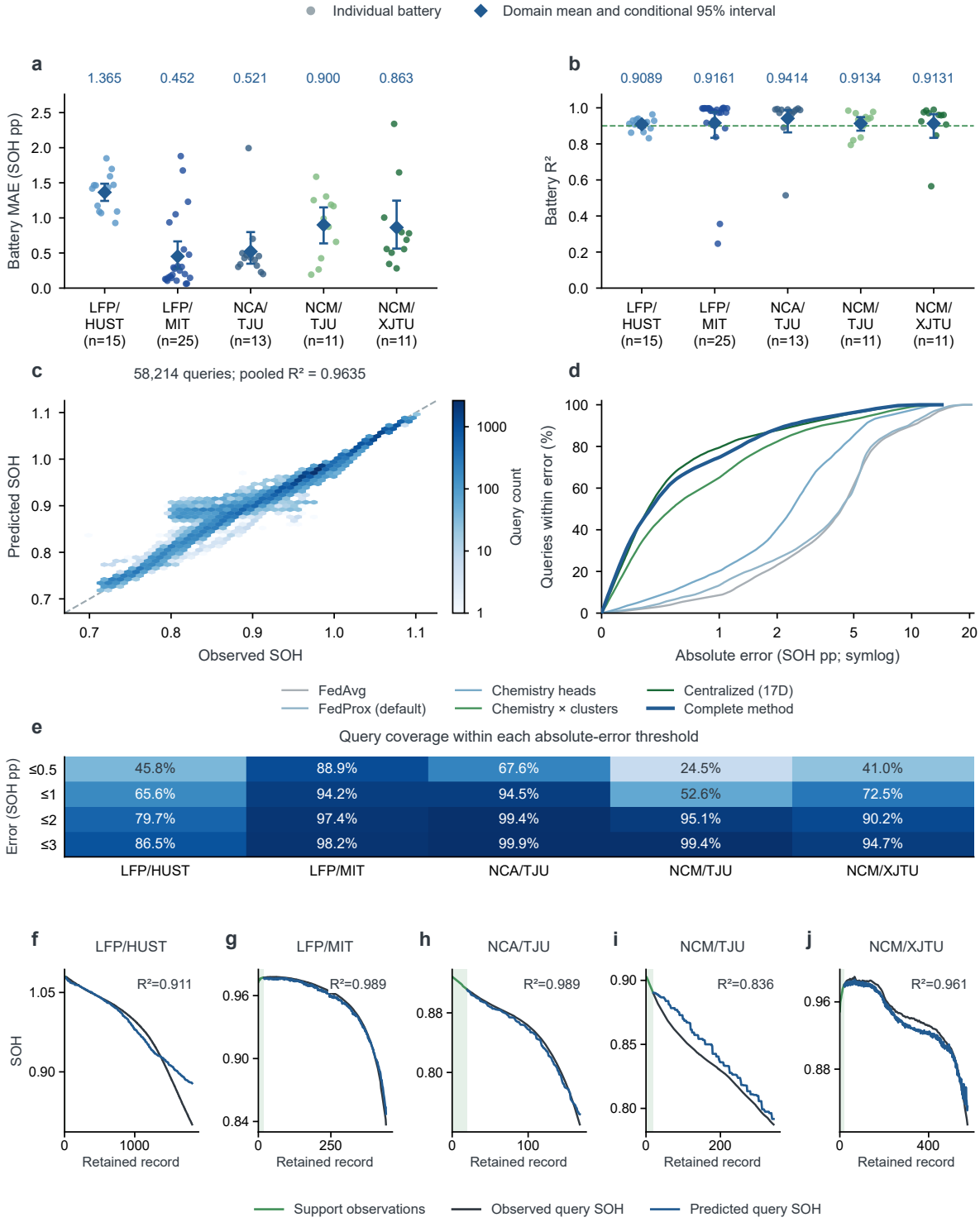

Individual battery
Domain mean and conditional 95% interval
a
1.365
0.452
0.521
0.900
0.863
Battery MAE (SOH pp)
LFP/ HUST (n=15)
LFP/ MIT (n=25)
NCA/ TJU (n=13)
NCM/ TJU (n=11)
NCM/ XJTU (n=11)
b
0.9089
0.9161
0.9414
0.9134
0.9131
Battery R²
c
58,214 queries; pooled R² = 0.9635
Predicted SOH
Observed SOH
Query count
d
Queries within error (%)
Absolute error (SOH pp; symlog)
FedAvg
FedProx (default)
Chemistry heads
Chemistry × clusters
Centralized (17D)
Complete method
e
Query coverage within each absolute-error threshold
Error (SOH pp)
≤0.5 45.8% 88.9% 67.6% 24.5% 41.0%
≤1 65.6% 94.2% 94.5% 52.6% 72.5%
≤2 79.7% 97.4% 99.4% 95.1% 90.2%
≤3 86.5% 98.2% 99.9% 99.4% 94.7%
LFP/HUST
LFP/MIT
NCA/TJU
NCM/TJU
NCM/XJTU
f
LFP/HUST
R²=0.911
g
LFP/MIT
R²=0.989
h
NCA/TJU
R²=0.989
i
NCM/TJU
R²=0.836
j
NCM/XJTU
R²=0.961
SOH
Retained record
Support observations
Observed query SOH
Predicted query SOH